HASSAN GHANAM

Software Engineer | Backend Developer

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PROFESSIONAL SUMMARY

Software Engineer from Damascus University, specializing in robust backend architecture and the engineering foundations of data systems. Focused on the core challenges of backend engineering—structuring clean domain layers, designing scalable RESTful APIs, and building high-performance data flows using PHP/Laravel and SOLID principles. Practical experience in Redis caching, asynchronous event processing, and custom Compiler Design architectures. Strong technical foundation in database administration, data engineering, and Information Retrieval / Vector Search Systems, backed by an advanced algorithmic and competitive programming background (Codeforces Pupil).

TECHNICAL SKILLS

- Languages & Core: PHP, C++, Java, JavaScript, SQL, HTML/CSS, Python
- Backend & Frameworks: Laravel (RESTful APIs, MVC, DTOs, Repositories, Events & Listeners, Service Layer)
- Architecture & Systems: Modular Monolith, Clean Architecture, SOLID Principles, Custom Compiler Design, Asynchronous Queues & Jobs, Caching (Redis), Rate Limiting, Load Balancing, SDLC & UML
- Data Engineering & Search Mechanics: MySQL, SQLite, Relational DB & ERD Design, ACID Compliance, Vector Databases (FAISS), Semantic Search & Embeddings, BM25 & Hybrid Search Fusion, IR Evaluation Metrics
- Tools & Practices: Linux, Docker, Postman API, JSON, Git, GitHub, CI/CD Pipelines, Agile/Scrum, Requirements Gathering, Data Structures & Algorithms, AI-Assisted Development & Agentic Workflows

KEY PROJECTS & EXPERIENCE

(EXPLORE ADDITIONAL PROJECTS ON MY PORTFOLIO / GITHUB)

Unified Digital Portal for Syrian Universities — Graduation Project **Dec 2025 - Aug 2026**
Laravel, PHP, Redis, MySQL, RESTful APIs, CI/CD

- Designed and engineered a robust backend system using PHP/Laravel, adhering to Clean Architecture and SOLID principles (Service & Repository patterns, DTOs, Custom Middleware).
- Architected an asynchronous event-driven pipeline using Laravel Queues & Background Jobs for heavy PDF report generation, solving complex Arabic text rendering, RTL typesetting, and performance bottlenecks.
- Implemented real-time, large-scale notification and announcement dispatchers targeting thousands of active students using Laravel Reverb and WebSocket architecture.
- Automated continuous integration and deployment workflows with CI/CD pipelines, maintaining clean RESTful API contracts and optimized query performance.

Information Retrieval Engine (IR-System) **Jun 2026 - Jun 2026**
Python, Flask, FAISS, NLTK, Sentence-Transformers, SQLite

- Engineered a modular Information Retrieval engine processing 382k+ documents (BEIR Touche-2020 dataset), integrating statistical (BM25 / VSM) and semantic search models using Sentence-Transformers (BERT).
- Fine-tuned BM25 hyperparameters (k1=2.0, b=0.60) and developed a Hybrid Search Fusion model (RRF / CombSUM), achieving peak retrieval performance at MAP@10 = 0.1470 and Precision@10 = 76.12%.
- Integrated FAISS vector database with L2 normalization and memory-mapped files (mmap), reducing BERT query latency from ~1.5s to ~0.1s.
- Built text preprocessing pipelines (NLTK stemming, Unicode normalization) and deployed a dynamic analytics dashboard using Python/Flask.

More projects including Custom Compiler, Game Engine Optimization, and other systems are available on my Portfolio.

EDUCATION

Bachelor's Degree in Information Technology Engineering **Sep 2021 - Sep 2026**
Specialization in Software Engineering & Information Systems
Damascus University, Syria